

Limited-Reference Image Quality Assessment: Paradigms and Discussions

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ABSTRACT

Image enhancement techniques play an important role in improving visual quality of media content. Image Quality Assessment (IQA) metrics are essential in comparing and improving image enhancement algorithms. Although these enhancement methods belong to different types of tasks in computer vision, the IQA for these methods has a common problem, namely, their reference information is limited. This poses a challenge to IQA: How to utilize the limited reference information to evaluate the qualities of distorted images. I am dedicated to approaching this challenge throughout my PhD. In this paper, first, I illustrate the motivation of my PhD project. Then, I formulate the problem and the ultimate goal of my PhD research. Next, I introduce some developed results in my research. Finally, I present my ongoing work and future directions.

CCS CONCEPTS

• Computing methodologies → Image processing.

KEYWORDS

Image Quality Assessment, Limited Reference, Unified Model, Image Enhancement.

ACM Reference Format:

Keke Zhang. 2023. Limited-Reference Image Quality Assessment: Paradigms and Discussions. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3581783.3613436>

1 INTRODUCTION

With the rapid developments of computer vision and display devices, the demand for high quality media content has been increasing rapidly. This fact has promoted the advances of various image enhancement technologies, such as image Super-Resolution (SR), image retargeting, Underwater Image Enhancement (UIE) and Low-Light Image Enhancement (LLIE), etc. The past few years have witnessed a booming of those techniques with massive applications, including remote sensing, medical image processing, image editing, marine environmental protection, road monitoring, and many others. Nevertheless, the qualities of enhanced images are

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MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

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ACM ISBN 979-8-4007-0108-5/23/10...\$15.00

<https://doi.org/10.1145/3581783.3613436>

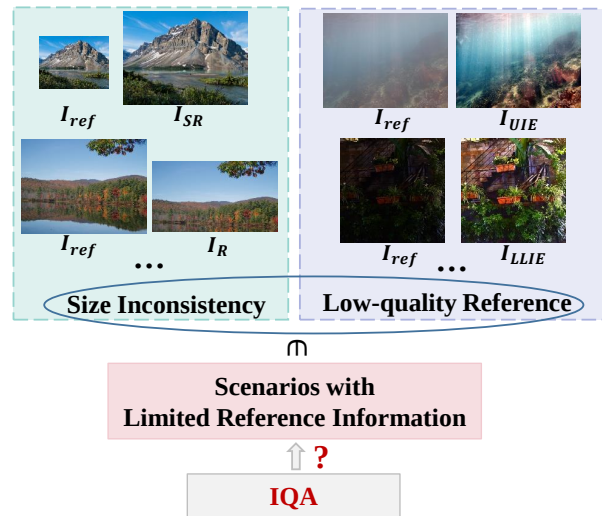


Figure 1: Illustration of the motivation of my PhD project. For image SR and retargeting tasks, there exist the size inconsistency problem between reference and distorted images. Towards UIE and LLIE tasks, their reference images are low-quality. The IQA for these tasks all belong to scenarios with limited reference information. My PhD project is dedicated to exploring a challenging problem: How to utilize the limited reference information to evaluate the qualities of distorted images?

not always visually pleasing. Quality variations may exist across different enhancement methods. To compare and optimize these enhancement algorithms, an Image Quality Assessment (IQA) metric is imperative.

Although these image enhancement techniques belong to different categories in computer vision community, the IQA for these applications have a common problem, that is, for a distorted image, its reference information is limited. I refer to this type of IQA as limited-reference IQA. Specifically, regarding image SR and retargeting tasks, the sizes of reference images are inconsistent with SR images and retargeted images. Towards UIE and LLIE tasks, their reference images are obviously low-quality. However, reference information is still indispensable to guarantee structural consistency in these enhancement processes. This poses a challenge to IQA: How to utilize the limited reference information to evaluate the qualities of distorted images (as shown in Fig. 1)? My PhD project is devoted to approaching this challenge.

Recent years have emerged numerous IQA methods[22], I divide these methods into task-specific and general methods according to their application ranges. Between them, task-specific IQA models generally only focus on the distortions introduced by a specific image processing task. Accordingly, these models have poor generalization ability and are inflexible in practice (see Section II for details). General IQA models are assumed to be able to handle various common distortions, including distortions introduced by acquisition, compression, transmission and so on [22]. Nevertheless, they often fall short when evaluating the qualities of distorted images in the scenarios with limited reference information (see Section II for details). Based on the above analyzes, I set the ultimate goal of my PhD project to develop a unified model for limited-reference IQA.

2 RELATED WORK

Existing IQA methods can be divided into subjective and objective methods. Between them, subjective tests are more reliable, but they are expensive and cannot be embedded in real-world applications [22]. Therefore, objective IQA methods are a research hotspot in IQA community. According to the availability of undistorted reference images, objective IQA metrics can be divided into referenced metrics, including Full-Reference (FR) and Reduced-Reference (RR), and No-Reference (NR) metrics. Further, each type of IQA methods can be classified into task-specific and general methods.

The situation of the reference image of the referred limited-reference IQA is shown in Fig. 1. The difference between it and FR-IQA is that the reference image of FR-IQA is undistorted and generally have the same size with distorted images. The difference between limited-reference IQA and RR-IQA is that the reference image of RR-IQA is also undistorted but only partial information can be accessed. The difference between limited-reference IQA and NR-IQA is that the reference image of NR-IQA is not available.

2.1 Task-Specific IQA Models

According to the focus of this paper, I will introduce recent developments on the following four IQA tasks.

SR-IQA Models. Existing SR-IQA methods can be classified into handcrafted-features-based and deep-learning-based methods. For the former, Jiang *et al.* [7] developed a Karhunen-Loève Transform based metric for SR images. Zhou and Wang [31] proposed a metric to measure deterministic and statistical fidelity for image SR. For deep-learning-based methods, Zhou *et al.* [30] designed a two-stream CNN that comprises two subcomponents for structure and texture SR images. Zhao *et al.* [28] designed a deep image SR quality model by employing a two-stream CNN. The inputs of the model are SR and LR images, respectively.

IRQA Models. Considering that fidelity is important in retargeting task, most IRQA methods are referenced [1]. Hsu *et al.* [5] designed an IRQA metric that consider both perceptual geometric distortion and information loss. Zhang *et al.* [26] proposed a backward registration-based aspect ratio similarity metric for IRQA. Based on [26], Zhang *et al.* [27] also designed a multiple-level feature-based quality measure for IRQA. Jiang *et al.* [6] designed a deep-learning-based model for IRQA that integrates geometric and content features. Li *et al.* [8] proposed an instance semantics-based metric for IRQA.

Underwater IQA Models. Most underwater IQA models are based on handcrafted features. Guo *et al.* [4] proposed a metric by employing physics prior of underwater optical imaging and relevant properties of Human Visual System (HVS). Zheng *et al.* [29] developed a fidelity metric by exploiting naturalness, sharpness, and structure indexes of enhanced underwater images in CIELab space. Different from the above measures, Wang *et al.* [17] presented a deep-learning-based IQA model that includes a bi-directional reference generation module and two distortion evaluation networks for chromatic and sharpness.

Low-Light IQA Models. The development of low-light IQA metrics is still in its infancy. Xiang *et al.* [19] proposed an NR-IQA metric, which considers brightness and texture features of low-light images. Zhai *et al.* [23] developed a measure by evaluating luminance enhancement, color rendition, noise and structure preserving of enhanced low-light images. Yang *et al.* [21] proposed a measure by extracting enhancement-aware features, structure-preserve features and colorfulness of enhanced low-light images.

Although the aforementioned works achieved promising results on specific tasks, they may not be able to obtain acceptable performance on other types of image processing tasks. The possible reason is that they only focus on the distortions introduced by a specific image processing task. As a result, task-specific IQA models have poor generalization ability and are inflexible in practice.

2.2 General IQA Models

I empirically divide general IQA models into handcrafted-features-based models and deep-learning-based models.

Handcrafted-Features-Based General IQA Models. Wang *et al.* [15] proposed the milestone IQA metric SSIM, which is based on the hypothesis that HVS is highly adapted to extract structural information from visual scenes. Given the great success of SSIM, several SSIM-based variants have been proposed, such as multi-scale SSIM [18] and information content weighted SSIM [16]. Subsequently, gradient-based IQA methods have been explored, such as [24], [9] [20]. Meanwhile, a series of general NR-IQA methods have been developed, among which natural scene statistics based methods are mainstream, including [11], [10] and [13].

Deep-Learning-Based General IQA models. Such IQA methods generally belong to FR and NR methods. For FR-IQA, Guo *et al.* [3] designed a bilateral-branch multi-scale image quality estimation network. Ding *et al.* [2] proposed a model that exhibits explicit tolerance to texture resampling by unifying structure and texture similarity. NR-IQA is a challenging topic due to the lack of reference information. Sun *et al.* [14] proposed a distortion graph representation learning framework for IQA. Zhang *et al.* [25] built continual learning for IQA, where a model learns continually from a stream of IQA datasets.

Despite recent advances, existing general IQA models are limited in given aspects. First, FR-IQA methods cannot handle the IQA for image resizing tasks. This is because those methods usually require a distorted image to has the same resolution as its reference image. Second, FR-IQA methods cannot cope with the IQA on tasks with low-quality reference images well, since they assume that reference images are unimpaired and of perfect quality. Third, NR-IQA models are not entirely reliable in these tasks, since reference information

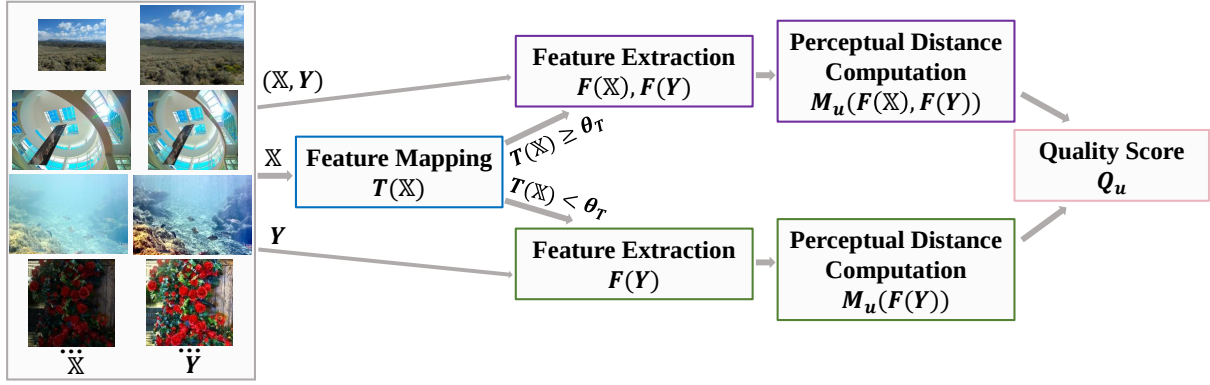


Figure 2: Illustration of my basic idea for the ultimate goal of my PhD project. The ideal unified IQA model comprises a feature mapping function $T(\cdot)$, a feature extraction module $F(\cdot)$ and a perceptual distance computation module $M_u(\cdot)$.

is still crucial to guarantee structural consistency during processing. As analyzed above, it is essential to design a unified IQA model for scenes with limited reference information.

3 PROBLEM DESCRIPTION

My PhD project is devoted to approaching the challenge: How to utilize the limited reference information to evaluate the qualities of distorted images. Further, the ultimate goal of my PhD project is to develop a unified IQA model to tackle this challenge. In addition, considering that reference information is indispensable to ensure structural consistency during image enhancement, this paper mainly focuses on referenced IQA models.

For traditional image processing tasks, such as image acquisition, compression, transmission, etc, there exist unimpaired reference images. Hence, the quality of a distorted image can be evaluated by a pixel-level similarity index with its reference image. This is a common operation in early general IQA metrics. The process can be expressed as:

$$Q_g = M_g(X, Y), \text{ s.t. } \text{size}(X) = \text{size}(Y) \wedge Q(X) = 1, \quad (1)$$

where X and Y represent a perfect reference image (labelled $Q(X) = 1$) and a distorted image, respectively. M_g denotes a general FR-IQA model. Q_g is the corresponding quality score.

However, for the scenarios with limited reference information (see Fig. 1), the above two constraints cannot be filled. Hence, the M_g cannot work in such cases. In view of this, researchers have developed a series of task-specific IQA models for different image enhancement tasks. This can be described as:

$$\begin{aligned} Q_{SR} &= M_{SR}(X_{LR}, Y), \\ Q_R &= M_R(X_{source}, Y), \\ Q_{UIE} &= M_{UIE}(X_{underwater}, Y), \\ Q_{LLIE} &= M_{LLIE}(X_{low-light}, Y), \end{aligned} \quad (2)$$

where Y implies a distorted image that can be an SR image, a re-targeted image, an enhanced underwater image and an enhanced low-light image. M_{SR} denotes an SR-specific IQA model, and Q_{SR} is the corresponding quality score, and so forth. As discussed above, task-specific IQA models have poor generalization ability and are inflexible in practice. Consequently, I am dedicated to developing a

unified IQA model for the scenarios with limited reference information. This can be represented as:

$$Q_u = M_u(\mathbb{X}, Y), \mathbb{X} \in \{X_{LR}, X_{source}, X_{underwater}, X_{low-light}\}. \quad (3)$$

Towards constructing the unified IQA model, I divide this challenging problem into two subproblems: First, whether to use the reference image? Second, how to use the reference image? In detail, for image SR and retargeting tasks, their reference images are unimpaired even though they have limited information. Accordingly, the extracted information/features from reference images are generally beneficial for quality assessment. However, for UIE and LLIE tasks, their reference images are impaired with varying degrees. In such case, the extracted information/features from severe damaged reference images may mislead the quality evaluation. Hence, it is reasonable for considering the first subproblem. Undoubtedly, the second subproblem is also crucial for designing IQA models.

To meet the first subproblem: Whether to use the reference image? I will analyze the information of the reference image first, and then adaptively adjust the unified IQA model M_u . Hence, the Eq. (3) can be revised to:

$$Q_u = \begin{cases} M_u(\mathbb{X}, Y), & \text{s.t. } T(\mathbb{X}) \geq \theta_T, \\ M_u(Y), & \text{s.t. } T(\mathbb{X}) < \theta_T, \end{cases} \quad (4)$$

where $T(\cdot)$ indicates a feature mapping function that aims at mapping a reference image $\mathbb{X}$ to a value. θ_T denotes a threshold. If $T(\mathbb{X}) \geq \theta_T$, the M_u will utilize the reference image. Otherwise, the M_u will not use the reference image.

To cope with the second subproblem: How to use the reference image? I will explore feature-level measurements to handle IQA for the scenes with limited reference information. Therefore, the Eq. (4) can be further adjusted to:

$$Q_u = \begin{cases} M_u(F(\mathbb{X}), F(Y)), & \text{s.t. } T(\mathbb{X}) \geq \theta_T, \\ M_u(F(Y)), & \text{s.t. } T(\mathbb{X}) < \theta_T, \end{cases} \quad (5)$$

where $F(\mathbb{X})$ and $F(Y)$ represent feature representations of $\mathbb{X}$ and Y , respectively. In a word, the ideal unified model consists of a feature mapping function $T(\cdot)$, a feature extraction module $F(\cdot)$ and a perceptual distance computation module $M_u(\cdot)$, as shown in Fig. 2.

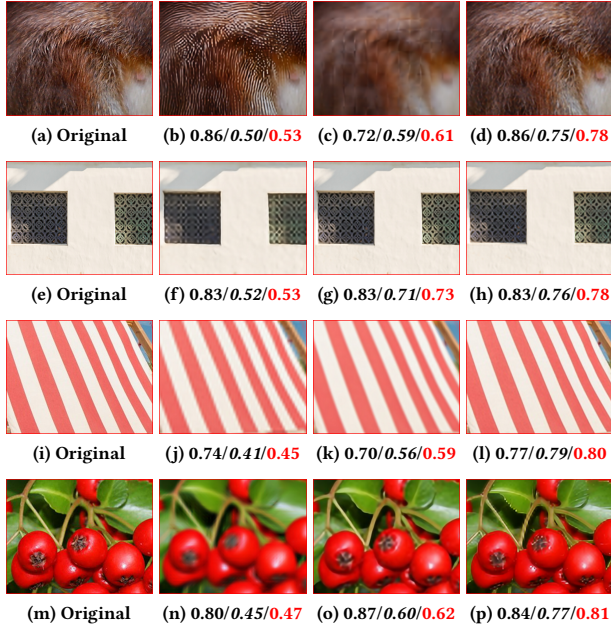


Figure 3: Visual comparison on the benchmark SISAR dataset [28]: DISQ($\uparrow$)/ours($\uparrow$)/MOS($\uparrow$). All images are cropped for better visibility.

4 RESULTS OBTAINED AND WORK IN PROGRESS

During the research of my PhD project, I have developed an SR-IQA metric, an IRQA metric and a preliminary unified IQA metric. With respect to the reproducibility of the proposed methods, I will release the codes after these works are officially accepted.

I show some visual comparison results of my SR-IQA metric and a state-of-the-art SR-IQA method DISQ [28] in Fig. 3. As can be observed, our SR-IQA metric has a lower prediction deviation compared with the DISQ model. The slight gap between our metric and Mean Opinion Score (MOS) indicates that our metric achieves superior performance in SR-IQA.

I show some visual comparison results of my IRQA metric and a state-of-the-art IRQA method MLF [27] on the benchmark dataset MIT RetargerMe in Fig. 4. Moreover, the subjective test used to the dataset is pairwise comparison, hence the label of each image is the votes it received. In such case, the performance of IRQA methods can be judged by the consistency with the ranking of votes. As can be observed in Fig. 4, our IRQA method outperforms the classic IRQA method MLF.

The designed preliminary unified IQA metric mainly considers the IQA for image SR and retargeting tasks. The IQA for these two tasks share a common characteristic that their reference images are unimpaired even with limited information. Hence I do not consider the first subproblem (see Section III for details) in designing the preliminary unified IQA model.

Based on my previous work, I am exploring the ultimate goal of my PhD project: Developing a unified model for limited-reference

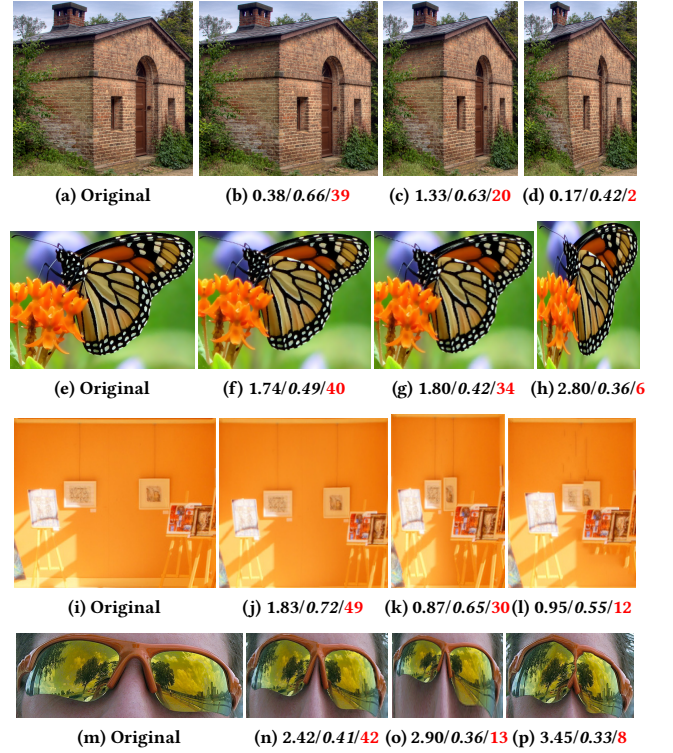


Figure 4: Visual comparison on the benchmark MIT RetargerMe dataset [12]: MLF($\uparrow$)/ours($\uparrow$)/votes($\uparrow$). All images are cropped for better visibility.

IQA. The problem formulation of the referred ultimate goal is presented in Section III. The key technologies consist of designing a feature mapping function $T(\cdot)$, a feature extraction module $F(\cdot)$ and a perceptual distance computation module $M_u(\cdot)$.

Regarding the feature mapping function $T(\cdot)$, which is correlated to a reference image $\mathbb{X}$ itself, I plan to consider the clarity and brightness of $\mathbb{X}$. For the feature extraction module $F(\cdot)$ and the perceptual distance computation module $M_u(\cdot)$, I plan to construct a deep-learning-based IQA framework to learn the two modules. Moreover, the ground truth of benchmark IQA datasets have different scoring ranges, which is also a problem when training various datasets simultaneously.

5 CONCLUSIONS

This paper presents an interesting and challenging problem throughout my PhD project, namely, how to utilize the limited reference information to evaluate the qualities of distorted images? This problem is common in IQA for image enhancement tasks. I present the problem formulation, ultimate goal, obtained results, ongoing work and future directions of my PhD project.

6 ACKNOWLEDGEMENTS

This paper was supported by Natural Science Foundation of Fujian Province, China (Grant No. 2022J02015).

REFERENCES

- [1] Bahareh Asheghi, Pedram Salehpour, Abdolhamid Moallemi Khiavi, and Mahdi Hashemzadeh. 2022. A comprehensive review on content-aware image retargeting: From classical to state-of-the-art methods. *Signal Processing* 195 (2022), 108496.
- [2] Keyan Ding, Kede Ma, Shiqi Wang, and Eero P Simoncelli. 2022. Image quality assessment: Unifying structure and texture similarity. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 5 (2022), 2567–2581.
- [3] Haiyang Guo, Yi Bin, Yuqing Hou, Qing Zhang, and Hengliang Luo. 2021. IQMA network: Image quality multi-scale assessment network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 443–452.
- [4] Pengfei Guo, Hantao Liu, Delu Zeng, Tao Xiang, Leida Li, and Ke Gu. 2022 (Early Access). An underwater image quality assessment metric. *IEEE Transactions on Multimedia* (2022 (Early Access)).
- [5] Chih-Chung Hsu, Chia-Wen Lin, Yuming Fang, and Weisi Lin. 2014. Objective quality assessment for image retargeting based on perceptual geometric distortion and information loss. *IEEE Journal of Selected Topics in Signal Processing* 8, 3 (2014), 377–389.
- [6] Bin Jiang, Jiachen Yang, Qinggang Meng, Baihua Li, and Wen Lu. 2020. A deep evaluator for image retargeting quality by geometrical and contextual interaction. *IEEE Transactions on Cybernetics* 50, 1 (2020), 87–99.
- [7] Qiuping Jiang, Zhentao Liu, Ke Gu, Feng Shao, Xinfeng Zhang, Hantao Liu, and Weisi Lin. 2022. Single image super-resolution quality assessment: A real-world dataset, subjective studies, and an objective metric. *IEEE Transactions on Image Processing* 31 (2022), 2279–2294.
- [8] Leida Li, Yixuan Li, Jinjian Wu, Lin Ma, and Yuming Fang. 2021. Quality evaluation for image retargeting with instance semantics. *IEEE Transactions on Multimedia* 23 (2021), 2757–2769.
- [9] Anmin Liu, Weisi Lin, and Manish Narwaria. 2012. Image quality assessment based on gradient similarity. *IEEE Transactions on Image Processing* 21, 4 (2012), 1500–1512.
- [10] Anish Mittal, Anush Krishna Moorthy, and Alan Conrad Bovik. 2012. No-reference image quality assessment in the spatial domain. *IEEE Transactions on Image Processing* 21, 12 (2012), 4695–4708.
- [11] Anush Krishna Moorthy and Alan Conrad Bovik. 2011. Blind image quality assessment: From natural scene statistics to perceptual quality. *IEEE Transactions on Image Processing* 20, 12 (2011), 3350–3364.
- [12] Michael Rubinstein, Diego Gutierrez, Olga Sorkine, and Ariel Shamir. 2010. A comparative study of image retargeting. In *ACM SIGGRAPH Asia 2010 papers*. 1–10.
- [13] Michele A Saad, Alan C Bovik, and Christophe Charrier. 2012. Blind image quality assessment: A natural scene statistics approach in the DCT domain. *IEEE Transactions on Image Processing* 21, 8 (2012), 3339–3352.
- [14] Simeng Sun, Tao Yu, Jiahua Xu, Wei Zhou, and Zhibo Chen. 2022. GraphQA: Learning distortion graph representations for blind image quality assessment. *IEEE Transactions on Multimedia* 25 (2022), 2912–2925.
- [15] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. 2004. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing* 13, 4 (2004), 600–612.
- [16] Zhou Wang and Qiang Li. 2011. Information content weighting for perceptual image quality assessment. *IEEE Transactions on Image Processing* 20, 5 (2011), 1185–1198.
- [17] Zheyin Wang, Liquan Shen, Zhengyong Wang, Yufei Lin, and Yanliang Jin. 2023. Generation-based joint luminance-chrominance learning for underwater image quality assessment. *IEEE Transactions on Circuits and Systems for Video Technology* 33, 3 (2023), 1123–1139.
- [18] Zhou Wang, Eero P. Simoncelli, and Alan C. Bovik. 2003. Multiscale structural similarity for image quality assessment. In *The Thirty-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, Vol. 2. 1398–1402.
- [19] Tao Xiang, Ying Yang, and Shangwei Guo. 2020. Blind night-time image quality assessment: Subjective and objective approaches. *IEEE Transactions on Multimedia* 22, 5 (2020), 1259–1272.
- [20] Wufeng Xue, Lei Zhang, Xuanqin Mou, and Alan C Bovik. 2013. Gradient magnitude similarity deviation: A highly efficient perceptual image quality index. *IEEE Transactions on Image Processing* 23, 2 (2013), 684–695.
- [21] Ying Yang, Tao Xiang, Shangwei Guo, Xiao Lv, Hantao Liu, and Xiaofeng Liao. 2023 (Early Access). EHNQ: Subjective and objective quality evaluation of enhanced night-time images. *IEEE Transactions on Circuits and Systems for Video Technology* (2023 (Early Access)).
- [22] Guangtao Zhai and Xiongkuo Min. 2020. Perceptual image quality assessment: A survey. *Science China Information Sciences* 63 (2020), 1–52.
- [23] Guangtao Zhai, Wei Sun, Xiongkuo Min, and Jiantao Zhou. 2021. Perceptual quality assessment of low-light image enhancement. *ACM Transactions on Multimedia Computing, Communications, and Applications* 17, 4 (2021), 1–24.
- [24] Lin Zhang, Lei Zhang, Xuanqin Mou, and David Zhang. 2011. FSIM: A feature similarity index for image quality assessment. *IEEE Transactions on Image Processing* 20, 8 (2011), 2378–2386.
- [25] Weixia Zhang, Dingquan Li, Chao Ma, Guangtao Zhai, Xiaokang Yang, and Kede Ma. 2023. Continual learning for blind image quality assessment. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 3 (2023), 2864–2878.
- [26] Yabin Zhang, Yuming Fang, Weisi Lin, Xinfeng Zhang, and Leida Li. 2016. Backward registration-based aspect ratio similarity for image retargeting quality assessment. *IEEE Transactions on image processing* 25, 9 (2016), 4286–4297.
- [27] Yabin Zhang, Weisi Lin, Qiaohong Li, Wentao Cheng, and Xinfeng Zhang. 2017. Multiple-level feature-based measure for retargeted image quality. *IEEE Transactions on Image Processing* 27, 1 (2017), 451–463.
- [28] Tiesong Zhao, Yuting Lin, Yiwen Xu, Weiling Chen, and Zhou Wang. 2022. Learning-based quality assessment for image super-resolution. *IEEE Transactions on Multimedia* 24 (2022), 3570–3581.
- [29] Yannan Zheng, Weiling Chen, Rongfu Lin, Tiesong Zhao, and Patrick Le Callet. 2022. UIF: An objective quality assessment for underwater image enhancement. *IEEE Transactions on Image Processing* 31 (2022), 5456–5468.
- [30] Wei Zhou, Qiuping Jiang, Yuwang Wang, Zhibo Chen, and Weiping Li. 2020. Blind quality assessment for image superresolution using deep two-stream convolutional networks. *Information Sciences* 528 (2020), 205–218.
- [31] Wei Zhou and Zhou Wang. 2022. Quality assessment of image super-resolution: Balancing deterministic and statistical fidelity. In *Proceedings of the 30th ACM International Conference on Multimedia*. 934–942.